Scientific Guide to Mushroom Cultivation

1. Introduction

- **Crop Name:** Oyster Mushroom (*Pleurotus spp.*)
- **Common Varieties:** *Pleurotus ostreatus* (White), *Pleurotus florida* (White), *Pleurotus sajor-caju* (Grey).
- **Nature:** It is a saprophytic fungus that grows on decomposing organic matter (lignocellulosic waste).
- **Nutritional Value:** High in protein (20-30% dry weight), fiber, and Vitamin B. Low in fat.

2. Environmental Requirements

- **Temperature:**
  - **Spawn Run (Vegetative Stage):** 24°C – 28°C
  - **Fruiting (Reproductive Stage):** 18°C – 24°C
- **Humidity:** High humidity is critical. Maintain **80% – 90%** relative humidity during fruiting.
- **Light:**
  - **Incubation:** Complete darkness.
  - **Fruiting:** Indirect light (200 lux) or dim tube light for 8–10 hours/day. Direct sunlight is harmful.
- **pH:** The substrate pH should be between **6.5 – 8.0**.

3. Substrate & Spawn

- **Substrate Material:** Paddy straw, wheat straw, chopped maize stalks, or sugarcane bagasse.
  - *Note:* Paddy straw is the most commercially viable substrate.
- **Spawn (Seed) Rate: 2% to 3%** of the wet weight of the substrate (approx. 200g spawn for 10kg wet straw).
- **Bag Size:** Polythene bags of size **12 x 24 inches** or **14 x 26 inches** (100–125 gauge thickness).

4. Cultivation Process (Step-by-Step)

Step 1: Substrate Preparation (Sterilization)

- Chop the straw into small pieces (2–3 inches).
- Soak in water for 4–6 hours.
- **Chemical Sterilization:** Dissolve **7g Bavistin** (Carbendazim) and **125ml Formalin** in 100 liters of water. Soak the straw in this solution for 18 hours.
- **Drying:** Drain excess water on a clean surface until moisture content is about **65%**. (Test: Squeeze a handful of straw; no water should drip, but the palm should feel wet).

Step 2: Spawning (Seeding)

- Layer Method: Put a layer of straw in the bag, then sprinkle spawn. Repeat for 4–5 layers.
- Close the mouth of the bag tightly with a rubber band.
- Make **8–10 small holes** with a needle for air exchange.

Step 3: Incubation (Spawn Run)

- Keep the bags in a **dark room** with proper ventilation.
- Maintain temperature at **25°C**.
- After **15–20 days**, the bag will turn completely white due to mycelium spread.

Step 4: Fruiting

- Once the bag is white, slash the polythene or remove it partially.
- Spray water (misting) on the walls and floor to increase humidity to **85-90%**.
- Allow fresh air circulation to reduce CO_2 levels.
- Tiny pinheads (baby mushrooms) will appear in 3–5 days.

5. Harvesting

- **Stage:** Harvest when the mushroom cap edges start to curl upwards but before they release spores.

- **Method:** Twist and pull the entire cluster from the base. Do not cut.
- **Yield:** A typical bag (4kg wet substrate) yields **2.5 – 3.0 kg** of fresh mushrooms in 3 flushes (harvest cycles).
- **Gap between Flushes:** 7–10 days.

6. Major Pests and Diseases

- **Green Mold (*Trichoderma spp.*):**
  - *Symptoms:* Green patches appear on the white mycelium.
  - *Cause:* Improper sterilization or unhygienic surroundings.
  - *Control:* Spray **Bavistin (0.1%)** on the affected area or discard the bag immediately.
- **Sciarid Flies (Mushroom Flies):**
  - *Symptoms:* Larvae eat the mycelium and tunnel into mushrooms.
  - *Control:* Install light traps; use nylon mesh on windows/vents.
- **Ink Caps (*Coprinus spp.*):**
  - *Symptoms:* Black, ink-like liquid oozes from the mushrooms.
  - *Cause:* High ammonia in substrate or old straw.
  - *Control:* Use fresh straw and proper pasteurization.